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AGLNet: Towards real-time semantic segmentation of self-driving images via attention-guided lightweight network

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Highlights

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AGLNet employs SS-nbt unit in encoder, and decoder is guided by attention mechanism.

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The SS-nbt unit adopts an 1D factorized convolution with channel split and shuffle operation.

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Two attention module, FAPM and GAUM, are employed to improve segmentation accuracy.

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AGLNet achieves available state-of-the-art results in terms of speed and accuracy.

Abstract

The extensive computational burden limits the usage of [convolutional neural networks](#) (CNNs) in edge devices for image [semantic segmentation](#), which plays a significant role in many real-world applications, such as [augmented reality](#), robotics, and self-driving. To address this problem, this paper presents an attention-guided lightweight network, namely *AGLNet*, which employs an encoder–decoder architecture for real-time semantic segmentation. Specifically, the encoder adopts a novel residual module to abstract feature representations, where two new operations, channel split and shuffle, are utilized to greatly reduce computation cost while maintaining higher [segmentation accuracy](#). On the other hand, instead of using complicated dilated convolution and artificially designed architecture, two types of [attention mechanism](#) are subsequently employed in the decoder to upsample features to match input resolution. Specifically, a factorized attention pyramid module (FAPM) is

used to explore hierarchical [spatial attention](#) from high-level output, still remaining fewer model parameters. To delineate object shapes and boundaries, a global attention upsample module (GAUM) is adopted as global guidance for high-level features. The comprehensive experiments demonstrate that our approach achieves state-of-the-art results in terms of speed and accuracy on three self-driving datasets: CityScapes, CamVid, and Mapillary Vistas. *AGLNet* achieves 71.3%, 69.4%, and 30.7% mean IoU on these datasets with only 1.12M model parameters. Our method also achieves 52 FPS, 90 FPS, and 53 FPS inference speed, respectively, using a single GTX 1080Ti [GPU](#). Our code is open-source and available at <https://github.com/xiaoyufenfei/Efficient-Segmentation-Networks>.

Introduction

Recently, building deeper and larger convolutional neural networks (CNNs) is a primary trend for solving robot vision tasks, such as image classification [1], [2], [3], object detection [4], [5], [6], and semantic segmentation [7], [8], [9]. To improve the representation ability of visual data, most accurate CNNs usually have hundreds even thousands of convolutional layers and feature channels, e.g., ResNet family [1], [10], [11]. In spite of achieving higher performance, these advances are at the sacrifice of running time and inference speed. Especially in the context of many real-world scenarios, such as augmented reality, robotics, and self-driving, the computationally cheaper networks with smaller model size are often required to carry out online estimation and decision in a timely fashion. Therefore, those state-of-the-art networks requiring enormous resources are not suitable for computationally limited mobile platforms (e.g., drones, robots, and smartphones), which have limited energy overhead, restrictive memory constraints, and reduced computational capabilities. This kind of limitation is particularly prominent on the computationally heavy task of semantic segmentation [8], [9], [12], [13], [14], which plays a significant role in robot vision [15], [16]. The goal here aims at helping robotics to understand surroundings by partitioning an input image into a series of disjoint regions, where each one is associated with a pre-defined semantic label including stuff (e.g., sky, road, buildings) and discrete objects (e.g., person, car, traffic light).

In order to adapt to real-world applications, many lightweight style convolutional neural networks have been designed to leverage the segmentation accuracy and implementing efficiency, which can be roughly divided into two categories: network compression [17], [18], [19], [20] and convolution factorization [21], [22], [23]. The first category prefers to reduce model size by compressing pre-trained networks, including hashing [17], pruning [18], and quantization [19], [20]. Quantization networks [19], [20] utilize less bits to encode model parameters instead of changing network architecture by pruning network weights [18]. To further remove the redundancy, an alternative approach to lighten CNNs depends on sparse coding theory [24], [25], where the model weights are always sparse, thus only small number of parameters are involved in inference calculation. On the contrary, motivated from the convolution factorization principle (CFP) [10], [22], [23], [26] that decomposes a standard convolution, the second category focuses on directly training a lightweight network from the perspective of reducing convolutional operation. For example, group convolution and depthwise separable convolution [2], [10] are widely used in MobileNet [23], [27] and ShuffleNet families [26], [28]. ENet [21] employs ResNet [1] as backbone to perform efficient inference. Zhao et al. [29] propose a cascade network that incorporates high-level label guidance to improve performance. In [13], [22], [30], an encoder-decoder architecture is adopted to restore object details, which greatly reduce the number of parameters while maintaining the accuracy. Although these advances have conducted preliminary research on designing lightweight architecture

networks for semantic segmentation, pursuing the best accuracy in very limited computational budgets still remains an open research problem for real-time semantic segmentation in robot vision.

In this paper, we aim at solving this trade-off as a whole, considering accuracy and run-time efficiency issues equally relevant. We introduce a novel lightweight network, called *AGLNet*, using an attention-guild encoder–decoder architecture for real-time semantic segmentation. As shown in Fig. 1, our *AGLNet* is composed of two parts: encoder and decoder network. Motivated from CFP [10], [22], [23], [26], the core unit of encoder is a computationally efficient residual module, adopting Split and Shuffle in a non-bottleneck structure (SS-nbt), that leverages identity mapping and 1-D factorized convolutions with channel split and shuffle. While the identity mapping allows the convolutions to learn residual functions that facilitate training, the split and shuffle operations enhance the information exchange within the feature channels while remaining similar computational costs compared to 1D factorized convolutions [22]. In the decoder, instead of using complicated dilated convolution [30], [31] and artificially designed architecture, two types of attention mechanism are employed to upsample features to match the resolution of input image. As illustrated in Fig. 1, a factorized attention pyramid module (FAPM) is used to extract dense features through exploring hierarchical spatial attention from low-resolution high-level output, where the factorized convolution is utilized to further lighten the entire network. Additionally, to delineate object shapes and boundaries, a global attention upsample module (GAUM) from low-level features is adopted as a global guidance for high-level semantics, in spite of adding a bit of computational burden, but achieving significant performance improvement. More specifically, the spatial attention is calculated from low-level features to assign weights for each pixel location of high-level features. Thereafter, channel attention are encoded to reweight channel features, where abundant category information can be used to select most important feature channels. Fig. 2 also shows the leaderboard of most recent light-weight networks for semantic segmentation on CityScapes dataset in terms of accuracy and efficiency. It can be seen that, although our method performs slower than ESPNet [30] and FPN [32], we improve 10% segmentation accuracy. Compared with DABNet [33] and EDANet [34], our approach reaches the best balance between accuracy and efficiency. In summary, the major contributions of our *AGLNet* are four-folds:

- $\hat{\phi}$

In contrast to previous lightweight networks [13], [22], [29], [30] that usually employ symmetrical architecture, our *AGLNet* adopts an asymmetrical encoder–decoder network structure. The encoder produces the downsampling features using SS-nbt unit, leading to the smaller network size while maintaining powerful representation ability. Subsequently, the decoder upsamples features with the guidance of attention mechanism, helping to improve segmentation accuracy while remaining high inference efficiency.

- $\hat{\phi}$

The SS-nbt unit of *AGLNet* adopts a split-transform-merge scheme to design residual layer, where 1D factorized convolution with channel split and shuffle operations leverages network parameters and feature representation, approaching the representational power of large and dense layers, but at a considerably lower computational complexity. In addition, channel shuffle is also differentiable, indicating that our *AGLNet* can be trained in an end-to-end manner.

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Two types of lightweight attention mechanisms, called FAPM and GAUM, are employed in decoder to improve segmentation performance. The hierarchical architecture of FAPM enlarges receptive fields of high-level features, allowing us to gather multiple scale context. On the other hand, the spatial and channel attentions are encoded in GAUM to aggregate the low-level spatial details and recalibrate the significance of feature channels efficiently.

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We test *AGLNet* on three self-driving datasets: CityScapes, CamVid, and Mapillary Vistas. The comprehensive experiments demonstrate that our approach achieves available state-of-the-art results in terms of speed and accuracy. Specifically, *AGLNet* achieves 71.3% and 69.4% mean IoU on the CityScapes and CamVid test set, and 30.7% on the Mapillary Vistas validation set, respectively, with only 1.12M model parameters and 52 FPS, 90 FPS, and 53 FPS inference speed on three datasets using a single GTX 1080Ti GPU.

The remainder of this paper is organized as follows. After a brief discussion of related work in Section 2, the detailed architecture of *AGLNet* is introduced in Section 3. The proposed network has been evaluated on three self-driving datasets: CityScapes [35], CamVid [36], and Mapillary Vistas [37], and the experiments can be found in Section 5. Finally, the concluding remarks and future work are given in Section 6.

Section snippets

Related work

In this section, we review the related advances for real-time semantic segmentation in robot vision using efficient CNN architecture.

AGLNet

In this section, we first introduce the core unit, SS-nbt, with split and shuffle operations in encoder. Thereafter, two proposed attention modules, FAPM and GAUM, are employed for semantic segmentation task. Finally, we describe the whole encoder-decoder network architecture of *AGLNet*.

End-to-end training of *AGLNet*

In training our *AGLNet*, one major problem is category unbalancing, where there is large variation in the number of training samples in each class. A typical example is the category of ‘‘road’’ and ‘‘road’’ in CityScapes dataset, in which the object instances of first class occupy a very small number of image regions, while those of the second one occupy a large number of pixels. Therefore, we train *AGLNet* in end-to-end manner using weighted cross entropy loss function. Let $z_k(x, \hat{I}_k)$ denotes

Experiments

This section reports the experimental results of our method on two challenging self-driving datasets: CityScapes [35] and CamVid [36]. Some ablation studies are also conducted to better understand the underlying behavior of our network for semantic segmentation in robot vision.

Conclusion remarks and future work

This paper has described a *AGLNet* model, which designs a lightweight encoder-decoder network for real-time semantic segmentation of self-driving images. The encoder adopts channel split and shuffle operations in residual layer, enhancing information communication in the manner of feature reuse. On the other hand, the decoder employs two attention blocks, FAPM and GAUM, where the first one adopts spatial pyramid architecture to enlarge receptive fields without introducing significant

CRedit authorship contribution statement

Quan Zhou: Conceptualization, Methodology, Writing - original draft. **Yu Wang:** Software, Validation, Investigation, Data curation. **Yawen Fan:** Writing - review & editing, Visualization. **Xiaofu Wu:** Writing - review & editing, Supervision. **Suofei Zhang:** Writing - review & editing, Project administration. **Bin Kang:** Writing - review & editing. **Longin Jan Latecki:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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2022, Computers and Electrical Engineering

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Automatic ultrasound image segmentation plays an important role in early diagnosis of human diseases. This paper introduces a novel and efficient encoder–decoder network, called Lightweight Attention Encoder–Decoder Network (LAEDNet), for automatic ultrasound image segmentation. In contrast to previous encoder–decoder networks that involve complicated architecture with numerous parameters, our LAEDNet adopts lightweight version of EfficientNet as encoder. On the other hand, a Lightweight Residual Squeeze-and-Excitation (LRSE) block is employed in decoder. To achieve trade-off between segmentation accuracy and implementing efficiency, we also present a family of models, from light to heavy (denoted as LAEDNet-S, LAEDNet-M, and LAEDNet-L, respectively), with varying lightweight version of EfficientNet backbones. To evaluate LAEDNet, we have conducted extensive experiments on Brachial Plexus Dataset (BP), Breast Ultrasound Images Dataset (BUSI), and Head Circumference Ultrasound Images Dataset (HCUS), where ultrasound images are suffered from high noise, blurred borders and low contrast. The experiments show that, compared with U-Net and its variants, e.g., M-Net, U-Net++ and TransUNet, our LAEDNet achieves better results in terms of Dice Coefficient (DSC) and running speed. Particularly, LAEDNet-M only has 10.75M model

parameters with 40.7 FPS, yet obtaining 73.0%, 73.8% and 91.3% DSC on BP, BUSI and HCUS datasets, respectively.

- **[CS-ResNet: Cost-sensitive residual convolutional neural network for PCB cosmetic defect detection](#)**

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In the printed circuit board (PCB) industry, cosmetic defect detection is an essential process to ensure product quality. However, existing PCB cosmetic defect detection approaches have a high false alarm rate, which lead to expensive labor costs of manual confirmation. To solve this problem, some traditional machine learning-based approaches have been proposed, but they just utilize hand-crafted features to build classifiers and thus are rough and sub-optimal. Recently, due to its powerful capability in automatic feature extraction, convolutional neural network (CNN) has been widely used in PCB cosmetic defect detection. However, few of them pay attention to the imbalanced class distribution as well as the different misclassification costs of real and pseudo defects, both of which are common problems in the PCB industry. To this end, in this study, we propose a novel model called cost-sensitive residual convolutional neural network (CS-ResNet) by adding a cost-sensitive adjustment layer in the standard ResNet. Specifically, we assign larger weights to minority real defects based on the class-imbalance degree and then optimize CS-ResNet by minimizing the weighted cross-entropy loss function. We conducted a series of experiments by comparing CS-ResNet with the standard ResNet, state-of-the-art CNN-based approach Auto-VRS and traditional machine learning-based approach HOG-SVM on a real-world PCB cosmetic defect dataset. Experimental results show that CS-ResNet achieves the highest Sensitivity (0.89), G-mean (0.91) and the lowest misclassification costs.

- **[DSANet: Dilated spatial attention for real-time semantic segmentation in urban street scenes](#)**

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Efficient and accurate semantic segmentation is particularly important in scene understanding for autonomous driving. Although Deep Convolutional Neural Networks(DCNNs) approaches have made a significant improvement for semantic segmentation. However, state-of-the-art models such as Deeplab and PSPNet have complex architectures and high computation complexity. Thus, it is inefficient for realtime applications. On the other hand, many works compromise the performance to obtain real-time inference speed which is critical for developing a light network model with high segmentation accuracy. In this paper, we present a computationally efficient network named DSANet, which follows a two-branch strategy to tackle the problem of real-time semantic segmentation in urban scenes. We first design a Semantic Encoding Branch, which employs channel split and shuffle to reduce the

computation and maintain higher segmentation accuracy. Also, we propose a dual attention module consisting of dilated spatial attention and channel attention to make full use of the multi-level feature maps simultaneously, which helps predict the pixel-wise labels in each stage. Meanwhile, Spatial Encoding Network is used to enhance semantic information and preserve the spatial details. To better combine context information and spatial information, we introduce a Simple Feature Fusion Module. We evaluated our model with state-of-the-art semantic image segmentation methods using two challenging datasets. The proposed method achieves an accuracy of 69.9% mean IoU and 71.3% mean IoU at speed of 75.3 fps and 34.08 fps on CamVid and Cityscapes test datasets respectively.

- [**LMFFNet: A Well-Balanced Lightweight Network for Fast and Accurate Semantic Segmentation**](#)

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